



Hello! How is it going?



good and what about you

fine

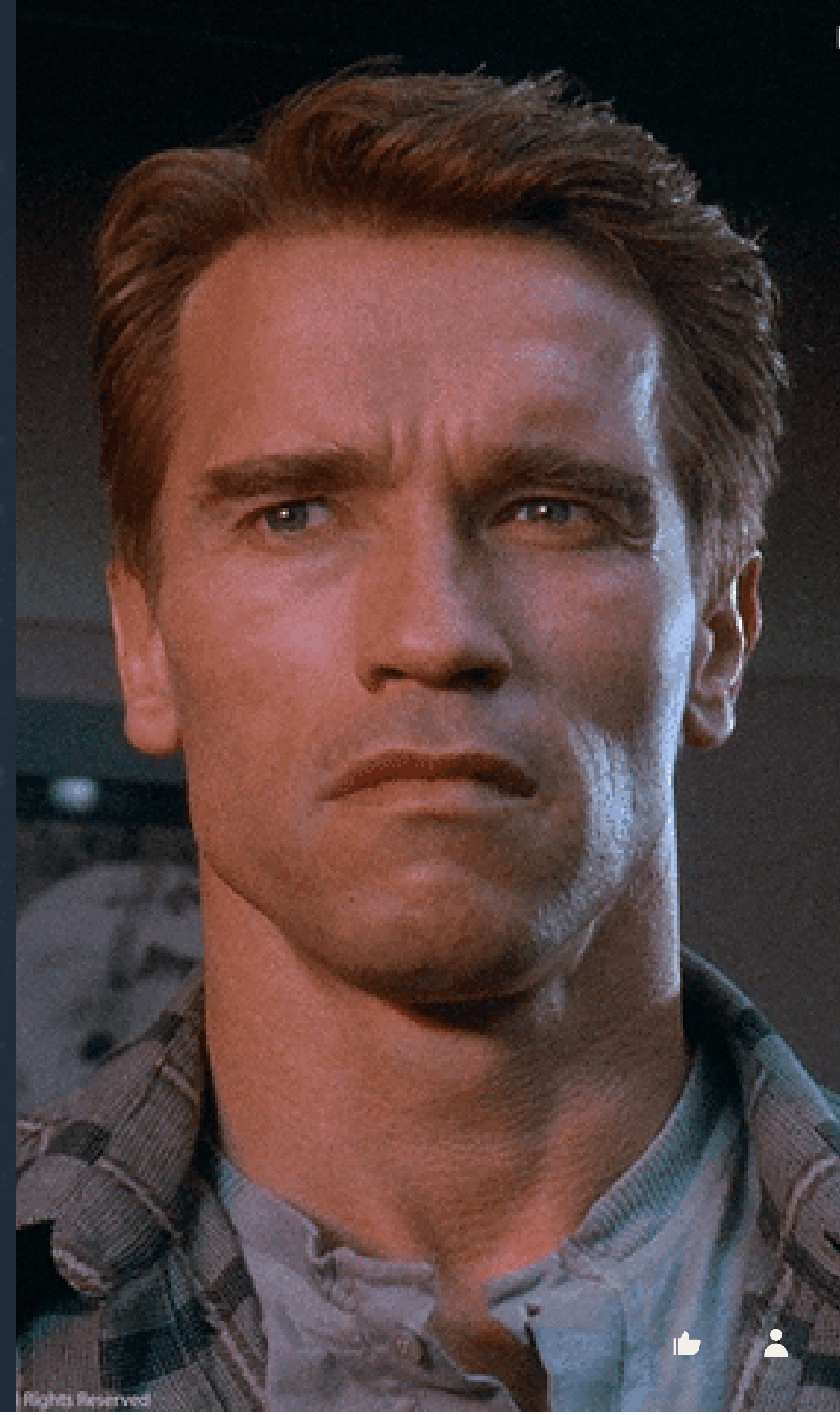
chatgpt refuses to work

not bad

Agenda

- Recall last session
- Discuss current topic
- Practice
- Quiz

Total Recall



What do you remember most from few previous workshops? (short answer)

json

java

xml

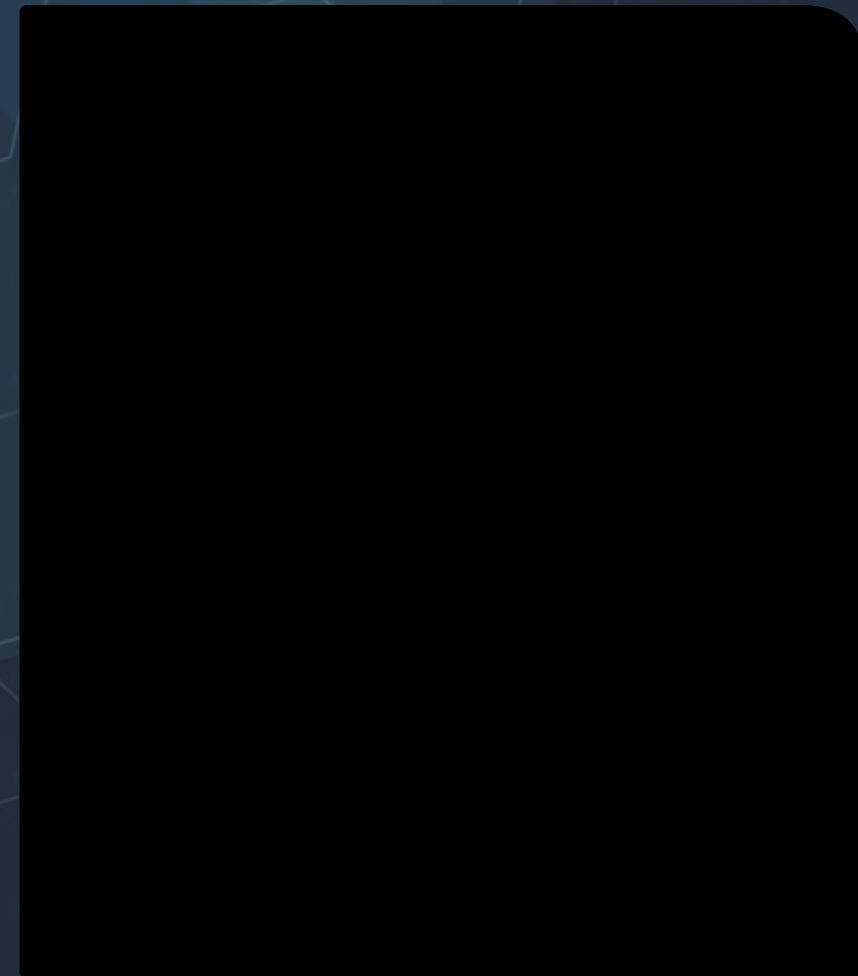
XML, Gson

jacskon gson

jdbc

Which Java library handles JSON parsing?

6 ✓



Jackson or Gson

1 ✗

XMLReader

1 ✗

JSONHandler

0 ✗

StringReader

Which format is XML stored in?

0 ✗

Binary

0 ✗

Tabular

1 ✗

Key-value pairs

5 ✓

Text and tree structure

What does an XML schema do?

0 ✕

Runs XML programs

1 ✕

Serializes the DOM tree

7 ✓

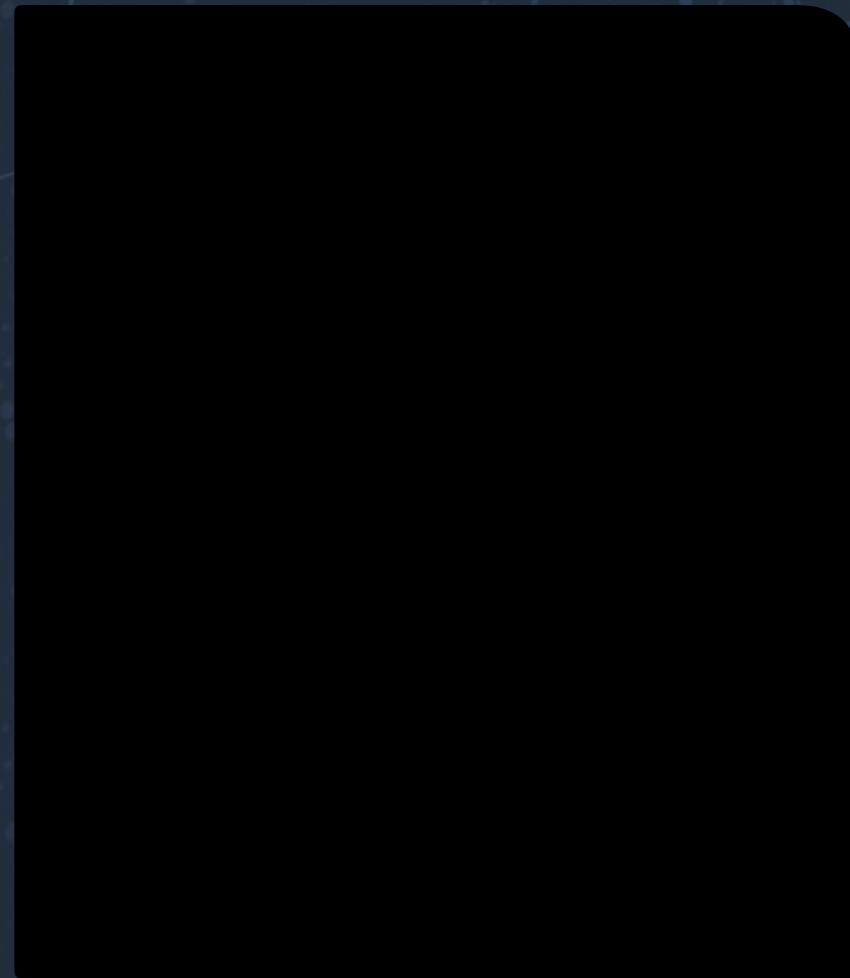
Defines rules for XML structure

0 ✕

Parses JSON objects

Example: What is the key difference between DOM and SAX?

7 ✓



0 ✗

SAX parses concurrently

0 ✗

DOM only reads XML

0 ✗

SAX uses attributes as data keys

DOM loads XML entirely, SAX
reads sequentially

Next topic

YOU!



Random fact about you

i've never been to africa

i removed ribs

i speak chinese i have extra ribs

my iq is 135 i am not a kazakh

my handgrip is 135kg

i broke bones and people'

i haven't slept in 36 hour wow

1 XML Parsers

Types of XML Processors

- Tree-based:
 - DOM - Document Object Model: The standard W3C tree interface built into modern web browsers and major programming languages.
 - etc.
- Streaming:
 - SAX (Simple API for XML): An event-driven, push-based streaming parser available in ecosystems like Apache Xerces.
 - StAX (Streaming API for XML): A pull-based streaming framework detailed by Oracle that allows the application to control parsing loops.
 - etc.
- Binding-style:
 - JAXB (Java Architecture for XML Binding): Is used to convert XML data directly into Java objects.
 - etc.

Tree-based Processors

- Represent entire XML document as object tree
- Can traverse tree forward or backward
- Pros: easy to search with XPath or API-specific tools
- Cons: more memory needed, slower than streaming

Streaming Processors

- Build or parse XML one node at a time
- Divided into "pull" and "push" processors
- Pros: fast and memory-efficient
- Cons: Complete data set not stored in memory; XPath searches not supported; SAX coding is complex

Push Streaming (SAX style)

- **Parser Controls Flow:** The XML parser reads the file and drives the execution.
- **Event Driven:** The parser generates events (e.g., **startElement**, **endElement**) as it encounters them.
- **Callback System:** The developer writes handler methods, and the parser "pushes" data into those handlers.
- **Passive Application:** The application must sit back, listen, and react to whatever the parser sends.

Pull Streaming (StAX style)

- **Application Controls Flow:** The developer's code drives the execution.
- **Cursor/Iterator Driven:** The application explicitly calls a method (like **.next()**) to request the next chunk of data.
- **Active Pulling:** The application "pulls" events out of the parser only when it is ready to process them.
- **Flexible Logic:** The application can stop parsing midway, skip sections, or easily c

Binding Processors

- Works like DOM-style in the background
- Developer annotates Java classes, runs simple code
- Pros: efficient programming model, easy to maintain
- Cons: JAXB doesn't work in Android
- Examples: JAXB, Simple

Comparison Table

Feature	DOM	SAX	StAX
Type	Tree-based	Push-streaming	Pull-streaming
Memory Usage	High (proportional to file size)	Low (constant)	Low (constant)
Parsing Speed	Slow (due to tree creation)	Fast	Fast
Navigation	Bi-directional (random access)	Forward-only	Forward-only
CRUD Operations	Read, Write, and Modify	Read-only	Read and Write
Control	Controlled by application	Controlled by parser	Controlled by application

BREAK



2 JAXB

Feature	JAXB
Type	Object-binding (maps XML directly to Java objects)
Memory Usage	High (proportional to the size of the generated object tree)
Parsing Speed	Medium to Slow (overhead from instantiation and reflection)
Navigation	Bi-directional (random access via the Java object model)
CRUD Operations	Read, Write, and Modify (unmarshal, marshal, and edit Java objects)
Control	Controlled by application (developer manipulates POJOs directly)

Key Architectural Differences

- **DOM vs JAXB:** DOM creates a generic tree of **Node** objects. JAXB creates a strongly-typed tree of specific domain classes (POJOs).
- **Streaming vs JAXB:** Unlike SAX and StAX, JAXB requires the entire XML document (or a massive chunk of it) to fit into memory as Java objects.



Quiz time

QUIZ

What does JAXB stand for?

16 ✓



0 ✗

Java Architecture for Binary Files

0 ✗

Java Application Binding Layer

0 ✗

Java-Array Binding API

Java Architecture for XML Binding

Which parser reads XML sequentially?

1 ✗

XSDParser

0 ✗

DOMParser

16 ✓

SAXParser

0 ✗

JSONParser

What is the purpose of Unmarshaller in JAXB?

18 ✓



0 ✗

Convert file to JSON

0 ✗

Create XML schema

0 ✗

Validate XML nodes

Which configuration is required for JAXB?

0 ✗

Schema validation

1 ✗

FileReader connection

1 ✗

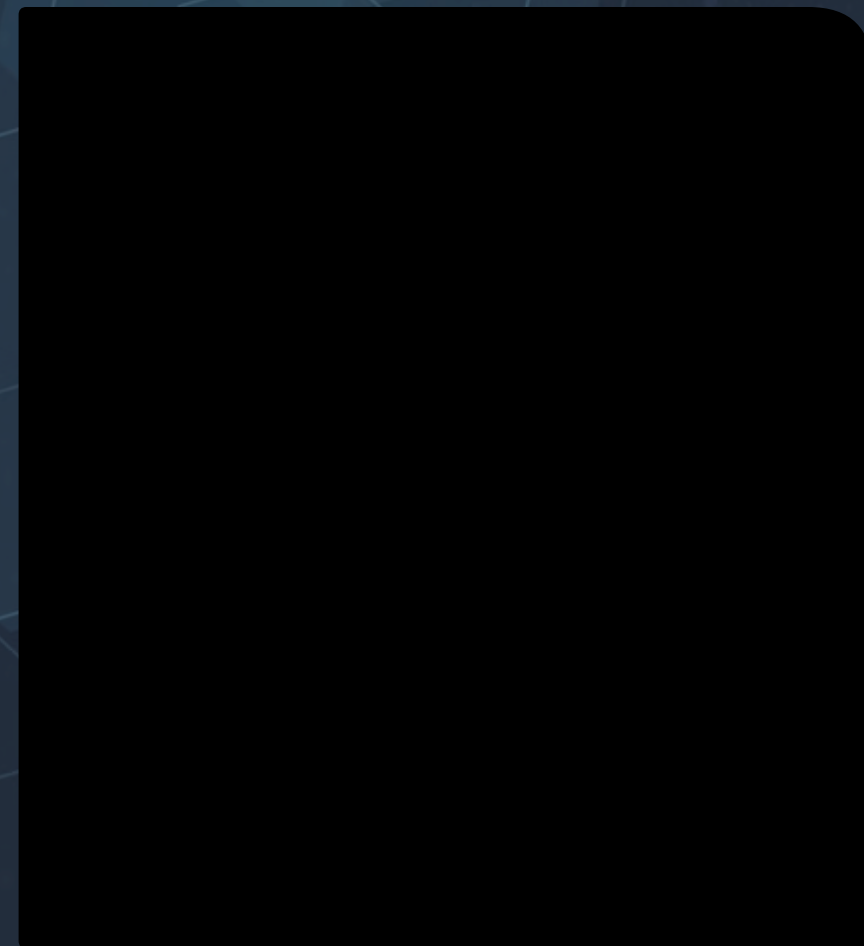
Class metadata override

15 ✓

Annotate classes with
`@XmlElement`

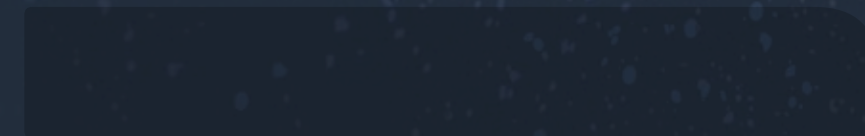
SAX parsing is best suited for?

16 ✓



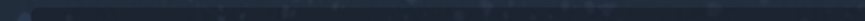
Low-memory sequential
reads

2 ✗



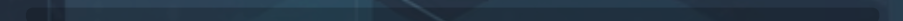
Analysis of complex
data

0 ✗



Edit and write large files

0 ✗



Object manipulations

What does @XmlElement indicate in JAXB?

18 ✓



0 ✗

Class metadata

0 ✗

Maps variable to XML element

Defines object type

0 ✗

Overrides SAX settings

What does DocumentBuilderFactory provide?

1 ✕

JSON object graphs

0 ✕

Abstract query templates

17 ✓

Interface for DOM-based parsers

0 ✕

Thread pooling management

What does the SAXParser work on?

0 ✗

Memory-intensive files

0 ✗

Handles performance
optimization

0 ✗

File objects in JSON format

18 ✓

Events generated during XML
reading

What does SAX handle directly?

0 ✗

Attributes as logs

0 ✗

Threaded reads

2 ✗

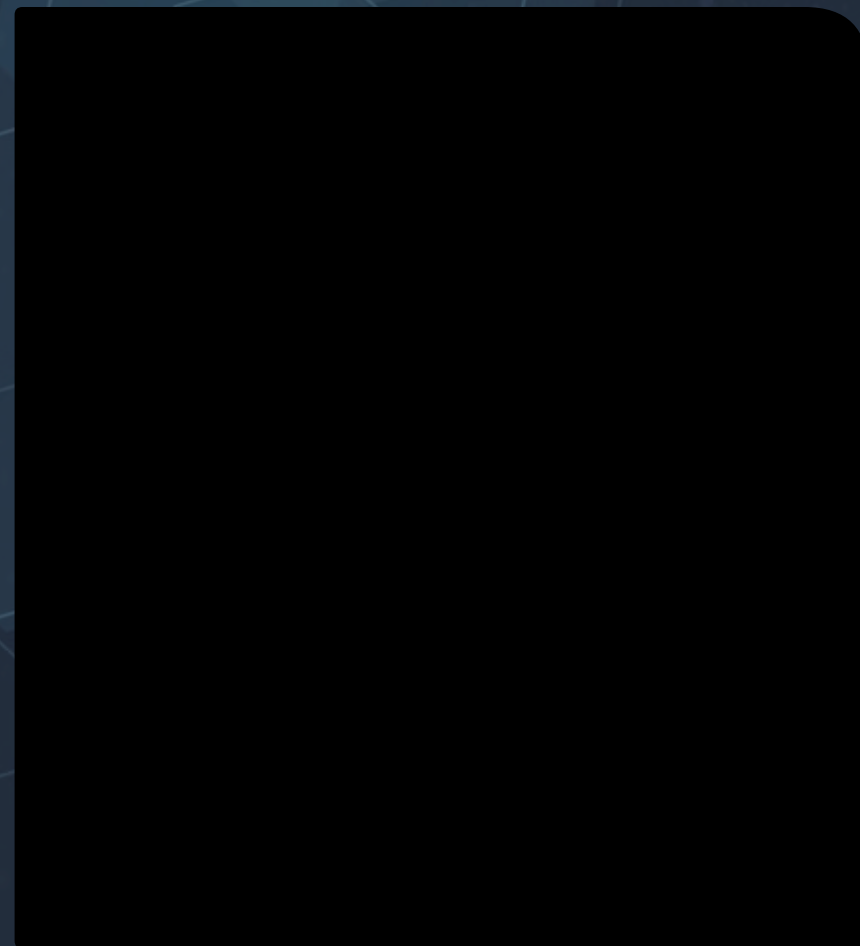
XML parsing exceptions

15 ✓

Tags and their data during traversal

What is XML binding in JAXB?

16 ✓



Convert between XML and Java objects

1 ✗

Only reading XML attributes

0 ✗

Writing HTML documents

0 ✗

Verify thread operations

Quiz leaderboard



Q&A part

0 questions
0 upvotes



